PHYC10004 Electromagnetism: Syllabus and lecture schedule

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Week	Beginning	Topics	Section	Text References[1]	Problems[1]	Lecture demonstrations
1	5 January	Electric charge	1	21.2, 21.3	21.24, 21.25, 21.28	El-3; El-1;
		Electric force and Coulomb's Law		21.1,21.4	21.5, 21.8, 21.13, 21.42	
		The electric field		22.1-6	22.26, 22.27, 22.31	El-5
		Electric dipoles		22.2, 22.7	22.19, 22.58	Balloons + 'confetti'; El-12; El-18
		Electric flux and Gauss' Law		23.1-2	23.1, 23.2, 23.3, 23.4	
		Gauss' Law - Examples		23.4-6	23.17, 23.22, 23.35, 23.38, 23.46	
		Electric fields and conductors		23.3	23.22, 23.35, 23.38, 23.46	El-13
		Electric potential energy	2	24.1		
		Electric potential		24.2	24.1, 24.4	
2	12 January	Electric potential – examples and applications		24.3-5, 24.7-8	24.13, 24.19, 24.22, 24.24, 24.45, 24.64	El-10; El-8; El-14
		Capacitance		25.1-2	25.2, 25.6, 25.7	El-4
		Energy storage in capacitors		25.4	25.32	
		Dielectrics		25.5-6	25.43, 25.52	El-4 (dielectric)
		Capacitors connected in series and parallel		25.3	25.11, 25.15	
		Electric current, Ohm's Law, and resistance	3	26.1-4	26.2, 26.5, 26.16, 26.28	Eg-2
		Power and electric circuits		26.5	26.48, 27.3	Ej-2
		Kirchhoff's Rules		27.1-2	27.10, 27.15, 27.21, 27.26, 27.29	
		Circuits with capacitors		27.3	27.57, 27.59	Ed-2
		Magnetic Fields - Introduction	4	28.1, 32.1, 32.4		Bar magnets+ compasses
		Magnetic force on a moving charged particle		28.1	28.4	Ef-2
		Magnetic & Lorentz force - Applications		28.2-5	28.8, 28.15, 28.25, 28.26	Eh-3
		Magnetic field from a current – Biot-Savart Law		29.1	29.3, 29.4, 29.6, 28.8	Ea-4; Ea-10
3	19 January	Ampére's Law		29.3-4	29.45, 29.43, 29.46	Ea-7
		Magnetic force on a wire carrying a current; Force and torque on a current loop		28.6-8, 29.2,29.5	28.44, 29.37	Ea-1; Ea-9; Ee-1
		Magnetic flux, induced currents and Lenz' Law		30.1	30.3, 30.5	Eb-2; Eb-9; Eb-11; Eb-10
		Faraday's Law		30.2	30.30, 30.36, 30.37	
		Faraday's Law – Applications and Examples		30.1-2		Ee-2; Eb-5; Eb-8; Eb-4
		Induced electric and magnetic fields		30.3, 32.2, 32.3	30.36, 32.2, 32.7, 32.15	
		Magnetic materials		32.1,32.5-8		Eh-1; Eh-4; Eh-6; Eh-10; Eh-8

[1] Walker et al, *Halliday's Fundamentals of Physics*, ANZ 1/e, Wiley, 2021